

# ME5412 Robotics for Healthcare

## Part 2

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# Course Instructor



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## Biorobotics Lab Overview



Construction, logistics,  
factories, and farming,  
caregivers



### Assisting Patients

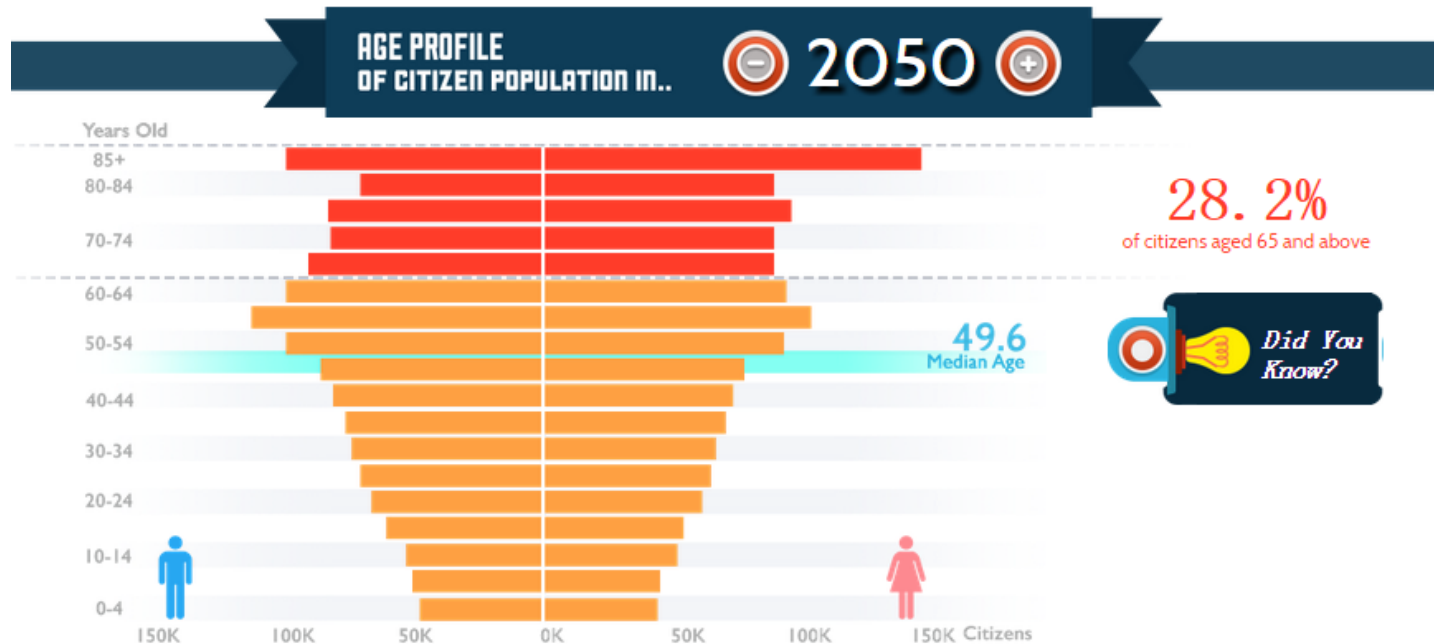
Stroke Rehabilitation  
Daily living assistance

*We develop novel robotic systems, devices, and enabling technologies to address the healthcare and manpower challenges in societies with increasing **aging population**.*



驱动力——人口老龄化

# Driving force: Aging Population



Source: Singapore Statistics

- 2050年新加坡公民人口年龄结构图（人口金字塔）
- 28.2% 的公民年龄在65岁以上
- 人口中位数：49.6岁



# 机器人学的复兴

## Resurgence of Robotics

- Demographic change
  - Ageing society
    - Escalating healthcare cost
  - Shrinking of young labor force
    - GDP growth from population increase to productivity gain
- Emerging new industry
  - AI and Robotics as rapidly growing industry

- 人口结构变化
  - 老龄化社会
  - 医疗成本攀升
  - 年轻劳动力萎缩
  - GDP增长动力从"人口红利"转向"生产率提升"
- 新兴产业
  - AI与机器人作为快速增长的产业

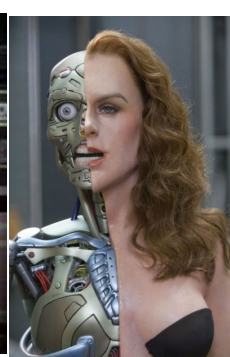
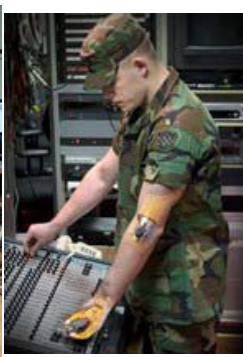
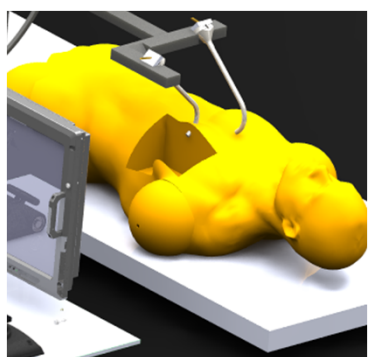


# 医疗领域的机器人

## Robotics in Healthcare

五大类：

- Surgical robots
  - Prosthetic robots
  - Rehabilitation robots
  - Assistive Robots
  - Social Robots
1. 外科手术机器人
  2. 假肢机器人
  3. 康复机器人
  4. 辅助机器人
  5. 社交机器人



手术模拟器、下肢外骨骼、军用康复臂、终结者机器人（拟人）、上肢康复机器人、轮椅搭载机械臂、社交机器人宠物、陪伴机器人

医疗机器人的应用领域

# Applications of Medical Robotics?

- 医疗机器人辅助临床任务：诊断、手术、康复
- Medical robots assist in clinical tasks such as diagnosis, surgery, and rehabilitation.
- Core areas include: • 核心领域：

- Surgical robotics: Enhancing precision and safety.
- Rehabilitation robotics: Supporting recovery from injuries or neurological conditions.
- Assistive robotics: Providing mobility and independence.
- Diagnostic robotics: Enabling faster and more accurate diagnoses.
- Social robotics: Offering companionship and mental health support.

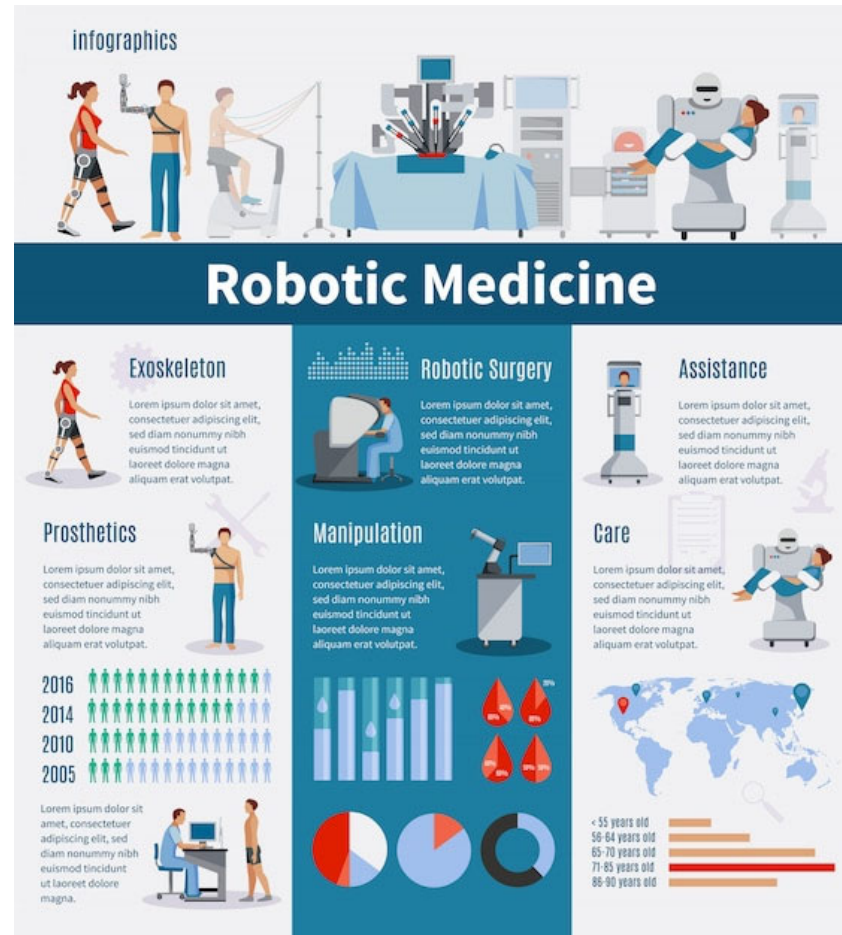
## ○ 外科机器人：提升精度与安全性

康复机器人：辅助损伤或神经疾病后的恢复

辅助机器人：提供行动能力与独立性

诊断机器人：实现更快、更准确的诊断

社交机器人：提供陪伴与心理健康支持

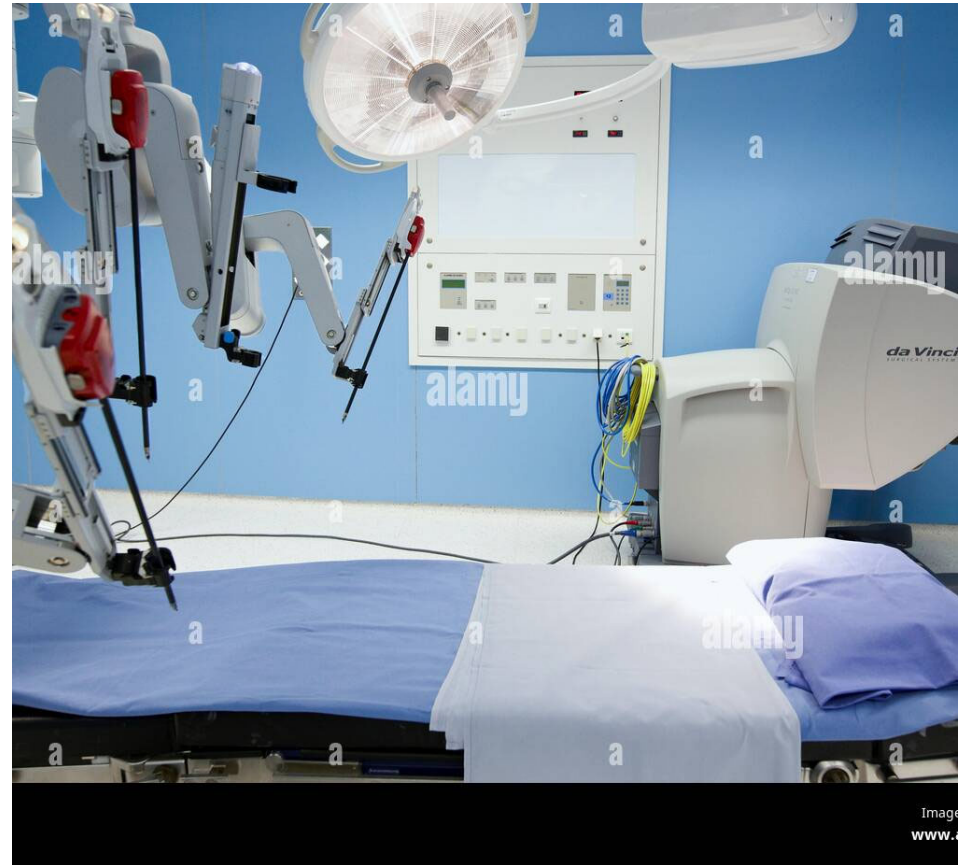


## 医疗机器人的演进

- 关键里程碑
  - 1990年代：
  - 2000年，达芬奇手术系统获FDA批准
  - MIT-Manus等康复机器人出现
  - 向儿科手术、肿瘤学等新领域扩展

## Evolution of Medical Robotics

- Key Milestones
  - 1990s:
    - FDA approval of the da Vinci Surgical System in 2000.
    - Emergence of rehabilitation robotics like MIT-Manus.
  - Expansion into new fields such as pediatric surgery and oncology.



达芬奇手术系统（da Vinci）手术室照片。



# Evolution of Medical Robotics

- Modern Developments
  - Integration of AI in diagnostics and navigation systems.
  - Use of cloud computing for data analysis and robotic learning.
  - Rise of wearable and portable robotic systems for home care.
- 现代发展
  - AI融入诊断与导航系统
  - 云计算用于数据分析与机器人学习
  - 可穿戴、便携式机器人系统兴起，用于居家照护



医生看CT影像

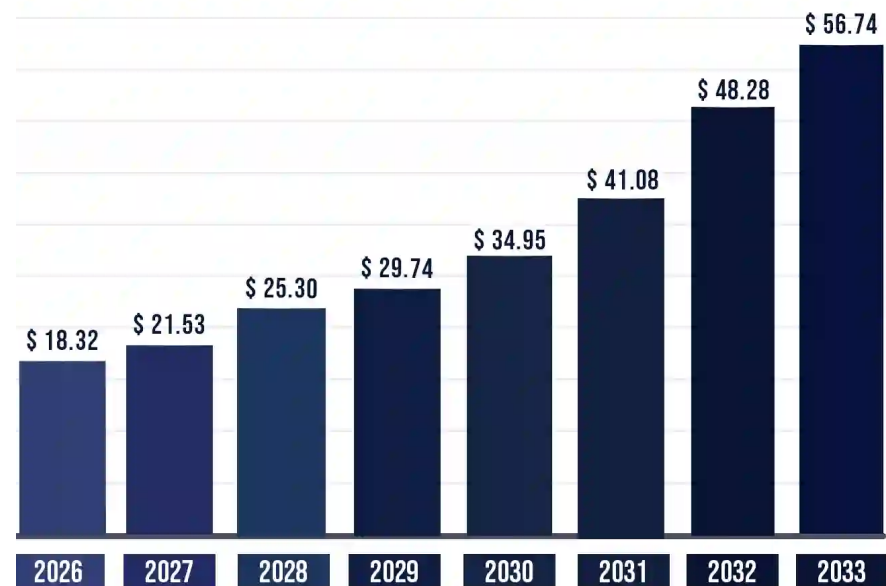


下肢外骨骼。

市场巨大，钱很多，所以这个方向值得做。

• 柱状图：2026年183.2亿 → 2033年567.4亿美元

## IBOTS MARKET SIZE 2023 TO 2033 (USD BILLION)



[www.precedenceresearch.com/medical-robots-market](http://www.precedenceresearch.com/medical-robots-market)



医疗机器人市场增长

## Market Growth in Medical Robotics

- Global market projected to reach \$50 billion by 2030.
- Key growth factors:
  - Aging population requiring advanced care solutions.
- Increasing demand for minimally invasive surgeries
  - 全球市场预计2030年达到500亿美元
  - 关键增长因素：
    - 老龄化人口需要先进护理方案
  - 微创手术需求增加

## 外科手术机器人

- 目前市场上有13个主要品牌的外科手术机器人

THERE ARE 13 MAJOR  
BRANDS OF ROBOTIC  
SURGERY UNITS ON THE  
MARKET TODAY

## Surgical robots

- 各种手术机器人器械图片（末端器械、医生控制台等）



达芬奇一家独大的时代正在过去，现在有13家竞品（如天智航、美敦力Mazor等）。

## 达芬奇手术系统

- 直观外科公司（Intuitive Surgical）开发
- 通过优越的视觉、增强的灵巧性、更高的精度和符合人体工学的舒适度，完成微创手术
- 切口仅1-2厘米，即可完成开胸手术等复杂操作
- 住院时间缩短一半，成本降低约三分之一（公司称）



达芬奇的核心价值是“微创”——医生不动手，操纵机械臂通过小孔进腹腔操作，3D视觉放大，手颤过滤。

## Da Vinci Surgical System

- The **Intuitive Surgical** company developed the **da Vinci robotic system** to perform minimally invasive surgeries through superior visualization, enhanced dexterity, greater precision, and ergonomic comfort. With incisions of only 1 or 2 centimeters, surgeons can perform even complex procedures such as open-heart surgery. The system reduces hospital stays by half, reducing costs by about one-third, because of less pain and speedier recovery, according to the company.



# Da Vinci Surgical System

- <https://www.youtube.com/watch?v=H4WzUSq9OeY>



# 康复工程

## Rehabilitation Engineering

系统地应用工程科学，为残障人士在功能领域（如行动、沟通、听觉、视觉、认知）以及与就业、独立生活、教育、社会融合相关的活动中所面临的问题，设计、开发、改造、测试、评估、应用和推广技术解决方案。

The systematic application of engineering sciences to **design, develop, adapt, test, evaluate, apply, and distribute technological solutions** to problems confronted by individuals with **disabilities in functional areas**, such as **mobility, communications, hearing, vision, and cognition**, and in **activities associated with employment, independent living, education, and integration into the community**.

*(USA's Rehabilitation Act of 1973)*

### Conventional Definition of rehabilitation engineering (RE)

康复工程（RE）的传统定义



这是本课程Part 2的正式定义——康复工程不是只做机器人，而是用一切工程手段帮残障人士回归社会。

# 辅助与康复装置

## Assistive and Rehabilitation Devices

- Elderly
- Disabled

### • 三点原则:

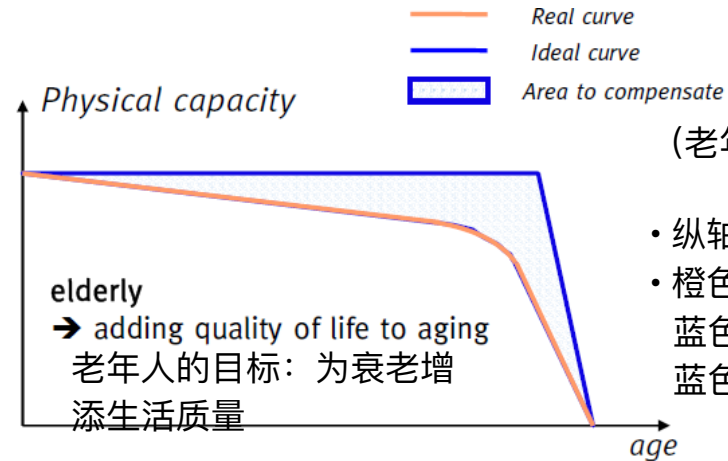
1. 尽早发现能力衰退
2. 补偿接近100%的能力
3. 尽可能长时间维持最高能力

### Physical capacity along time

- 1: detect the failing as soon as possible
- 2: compensate close to the 100% of capacity
- 3: continue the maximum of capacity as long as possible

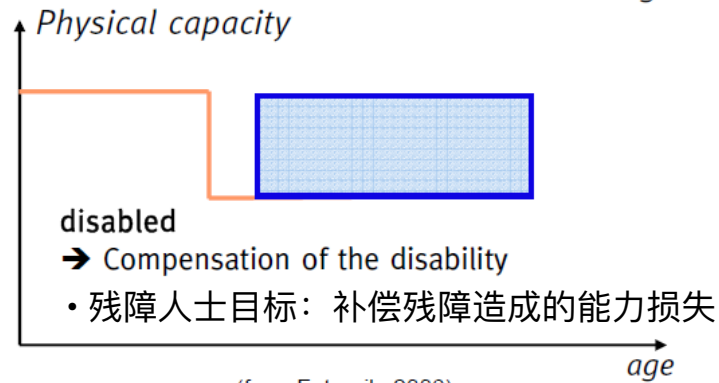
Goal of AT: provide tools to improve or **restore** physical performance

- 辅助技术 (AT) 的目标: 提供工具来提升或恢复身体表现



(老年人):

- 纵轴: 身体能力; 横轴: 年龄
- 橙色实线: 实际能力曲线;
- 蓝色虚线: 理想能力曲线;
- 蓝色阴影: 需要补偿的区域



(from Fatronik, 2006)

(残障人士):

- 身体能力在某个时间点突然下跌 (如中风、事故)
- 蓝色阴影区域: 需要补偿的缺口



# 什么是康复

## What is Rehabilitation

**Rehabilitation** is the process of helping an individual achieve the **highest level of function, independence, and quality of life** possible, with the aim of **integrating the individual into society**

康复是帮助个体达到最高水平的功能、独立性和生活质量的过程，目标是让个体融入社会

**Rehabilitation** does **not reverse or undo** the **damage** caused by **disease or trauma**, but rather helps **restore the individual to optimal health, functioning, and well-being**

康复不是逆转或消除疾病/创伤造成的损伤，而是帮助个体恢复到最佳健康、功能状态和幸福感



**Robotic Rehabilitation Center  
Riley Hospital for Children, Indianapolis**



关键概念——康复不“治愈”损伤，而是帮你在现有身体条件下做到最好。这是机器人康复的理论基础。

# 康复工程师

## Rehabilitation engineers

IBOT  
轮椅



### 参与领域 (Areas of involvement)

- 日常生活活动 (ADLs)：吃饭、如厕、梳头、刷牙、阅读
- 行动能力解决方案：
  - 个人行动：家/办公室内移动
  - 公共行动：汽车、公共交通、建筑无障碍
  - 在环境中有效地移动
  - 在特定环境中够取物体
  - 限制行动的障碍：不合适的轮椅坐垫或支撑系统
- 感知障碍解决方案：
  - 视觉、听觉、沟通障碍（输出端和理解端）

### 角色 (Roles)

- 通常在团队中工作
- 与治疗师、骨科医生、物理医学专家、神经科医生协作

### Roles

- Generally work in a team setting
- Collaboration with therapists, orthopedic surgeons, physical medicine specialists, neurologists

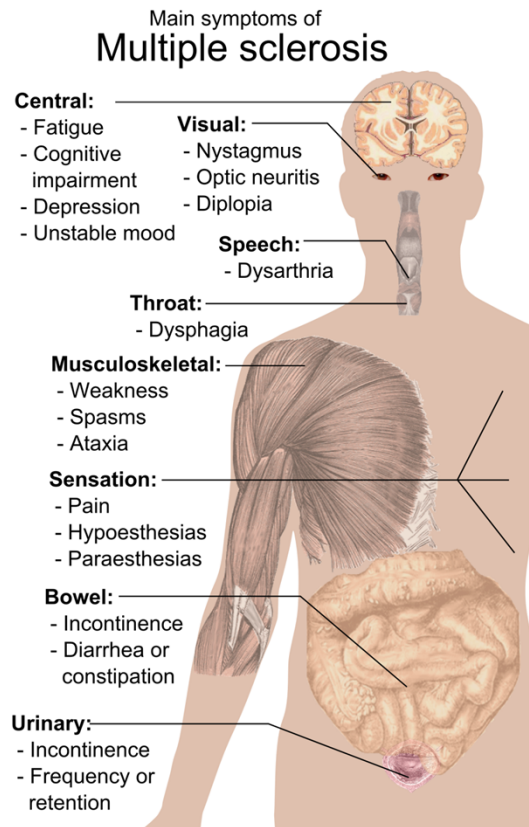
### Areas of involvement

- ADLs, e.g. eating, toileting, combing hair, brushing teeth, reading
- Solutions to mobility
  - *Personal mobility: within home/office*
  - *Public (automobile, public transport, accessibility issues in buildings)*
  - *Ability to move functionally through environment*
  - *Reaching for object in a particular setting*
  - *Barriers that limit mobility: ill-fitted wheelchair cushion or support system*
- Solutions to sensory disabilities
  - *Sight, hearing, communications disorders (in both production side and comprehension side)*



# 身体残障

## Physical Disability



• **医学疾病:** 中风、糖尿病、肌营养不良、关节炎、脑瘫、多发性硬化症.....

• **事故:** 截肢、脊髓损伤、脑损伤

## Medical conditions

- Stroke, diabetes, muscular dystrophy, arthritis, cerebral palsy, multiple sclerosis..

## Accidents

- Loss of limb
- Spinal cord injury
- Brain injury



*Motor neuron disease*



# 辅助技术的常见领域

## Common Areas of Assistive Technologies

假肢装置/人工肢体



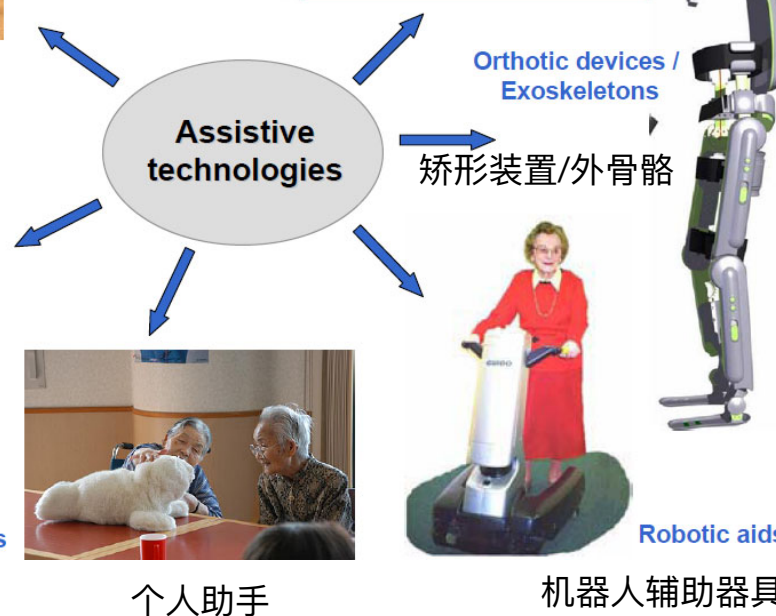
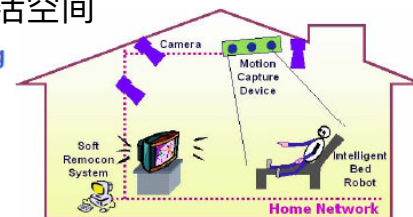
Prosthetic devices /  
Artificial limbs



FES  
FES功能性电刺激  
Personal assistants

智能生活空间

Smart living  
spaces

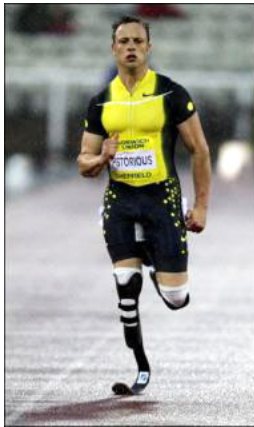


# 肢体缺失

## Loss of Limb

- 人工肢体设计

- 残肢越短（缺失越多），工程复杂度越高，尤其是丢失了一个或多个关节时
- 小腿截肢者用简单木腿就能走得不错



### Artificial limb design

- Engineering complexity increases as amount of residual limb decreases, especially if one or more joints are lost
- Person with transtibial amputation could use simple wooden stump to extend leg, and could ambulate reasonably well

### Advances in Prosthetics

- After WWII, concerted effort to foster better engineering design for artificial limbs in US
- **Dynamically lockable knee joints** were designed for above-knee amputees
- **Energy-storing artificial foot, Intelligent prosthesis**

- 假肢进展

- 二战后美国大力推动人工肢体工程设计
- 为大腿截肢者设计了动态可锁膝关节
- 储能脚、智能假肢



截肢位置越高（越靠近躯干），要代偿的自由度越多，假肢越难做。小腿截肢容易，大腿截肢难，肩离断/上臂截肢最难。

# 上肢假肢

## Upper Limb Prosthetics



# 上肢假肢 Upper Limb Prosthetics

## Passive/cosmetic

- 被动/装饰性
  - 外观极其自然
  - 维护难、昂贵
- Look extremely natural
- Difficult to maintain, expensive
- Sacrifice function for cosmetic appearance
- 功能性

## Functional

- 身体动力假肢（缆绳控制）
- 开关控制假肢
- 肌电假肢
- Body-powered prostheses (Cable-controlled)
- Switch-controlled prostheses
- Myoelectric prostheses

## Current research efforts

- Control of simultaneous multi-joint movement and provision for sensory feedback
- From end-effector of artificial arm to stump skin via electrical stimulation
- 当前研究方向
  - 多关节同时运动控制
  - 提供感觉反馈
  - 从人工手臂末端到残端皮肤的电刺激反馈



Jesse Sullivan



现在最大的研究痛点是：假手能抓，但使用者“感觉不到”手里拿了什么、用了多大力——需要把触觉反馈传回人体。

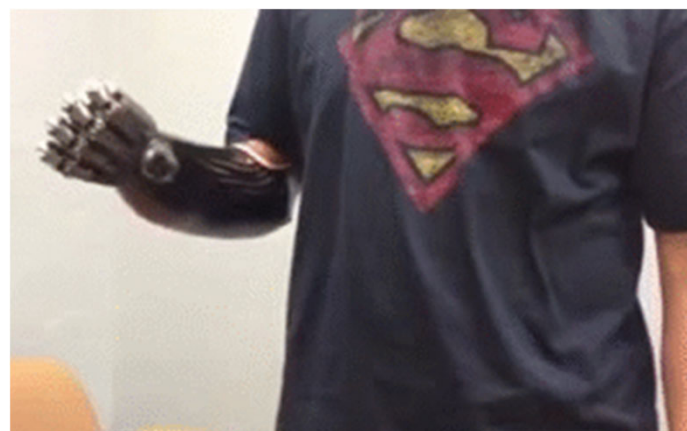
# 动力上肢假肢

## Powered Upper Limb Prosthetics

- 类型：肌电（智能）上肢假肢
- Types (Myoelectric (smart) upper-limb prosthetics)



(Myo prosthetics from Johns Hopkins )  
Johns Hopkins的肌电假肢



(Achilles prosthetics)  
Achilles假肢

- **Advantages:** improve functionality, adaptability, potential for continuous improvement
- **Disadvantages:** *cost, weight, complexity, usage time*
  - 优点：提升功能性、适应性、可持续改进的潜力
  - 缺点：成本高、重量大、复杂、使用时间短（用户不爱戴）



肌电假肢用残肢肌肉的电信号（EMG）来控制手的张开握紧。但现实很骨感——贵、重、难用，很多用户最后还是把假肢放家里不用。

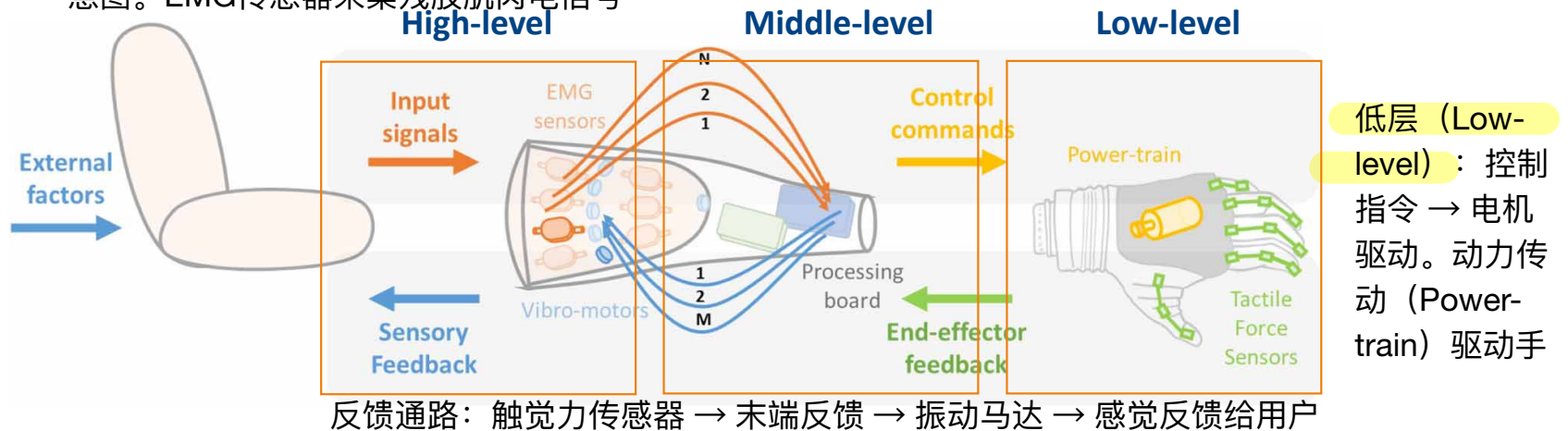
# 智能上肢假肢的控制

## Control of smart upper-limb prosthetics

- Hierarchical architecture of biological signal control 生物信号控制的分层架构

• 高层 (High-level) : 输入信号 → 运动意图。EMG传感器采集残肢肌肉电信号

中层 (Middle-level) : 运动意图 → 控制指令。处理板 (Processing board) 解码, 输出控制命令



[Progress in Biomedical Engineering, 2022]

- Hierarchical Architecture of biological signal control: lower-level control: from control commands to motor actuation; middle-level control: from movement intention to control commands; high-level control: from input signals to movement intentions.



# 动力假肢的挑战

## Challenges for powered prosthetics

这是一个根本性矛盾——截肢越高，假手要做的事越多，但你能用来控制的“信号源”越少。上臂截肢者肩膀只剩一点肌肉，却要控制整条假臂十几个自由度。

- **User interface and control**

- The human arm/hand is a complicated, high DOF system
- Emphasis is on deducing user intent from motor pathway (muscles, nerves, brain) for natural control
- Fundamental control problem for prosthetics:
  - The higher the limb is amputated, more DOF need to be controlled by a prosthetic
  - But more of the limb nerve and muscle is missing, offering less control input

- 用户界面与控制

- 人手臂/手是一个复杂、高自由度的系统
- 重点是从运动通路（肌肉、神经、大脑）中推断用户意图，实现自然控制
- 假肢的根本控制难题：
  - 截肢位置越高，假肢需要控制的自由度越多
  - 但同时，肢体的神经和肌肉丢失得越多，能提供的控制输入越少



# Challenges for powered prosthetics

## Motor pathway (muscles, nerves, brain)

运动通路（肌肉、神经、大脑）

Motor Cortex

1. 运动皮层（Motor Cortex）：大脑发出运动指令

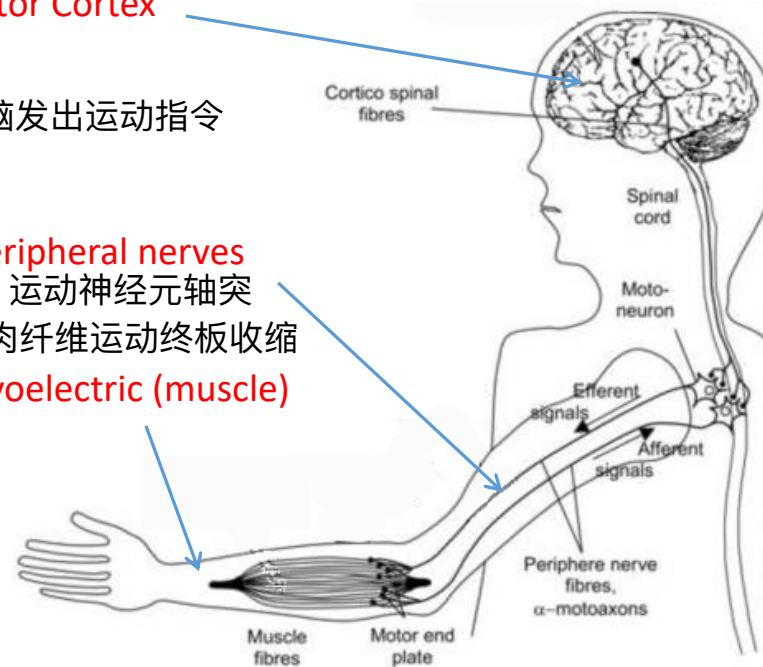
Peripheral nerves

3. 外周神经（Peripheral nerves）：运动神经元轴突

4. 肌电（Myoelectric / 肌肉）：肌肉纤维运动终板收缩

Myoelectric (muscle)

2. 脊髓：皮质脊髓束传导



解释：假肢控制就是在这条通路的不同节点“偷听”信号——在大脑（脑机接口）、在神经（外周神经接口）、在肌肉（EMG）。越往上读信号越丰富，但侵入性越大、手术越危险。

# 基于EMG的假肢控制

## EMG Based Control of Prosthetic Devices

- EMG的吸引力
  - 电极可贴在肌肉表面皮肤上

- Appeal of EMG
  - 风险极低、无创、可摘除
  - Electrodes may be placed on surface of skin over muscle
  - Minimal risk, noninvasive, removable
  - Well-understood concept ◦ 概念成熟、易理解

- How can a multi-DOF prosthetic be controlled with limited remaining musculature?

- Sharing control between user input and automation
  - (Manus hand)
- Moving motor neurons to different muscles
  - (Kuiken 2006)

- 核心问题：残肌有限，怎么控制多自由度假肢？

- 用户输入与自动化之间共享控制权（如Manus手）

- 把运动神经重新分配到不同肌肉（Kuiken 2006）



- **Myoelectrical-interface testing:** This testing detects the electrical signals from the residual muscles and evaluates the EMG activation mode for control of the smart upper-limb prosthetics.

肌电接口测试——在残肢肌肉上贴电极，评估EMG激活模式来控制智能上肢假肢。



EMG是目前最实用的方案（无创），但通道数有限。解决思路：一是让机器人自己“智能”一点（共享控制），二是做手术把神经重新接到更多肌肉上（下一页TMR）。

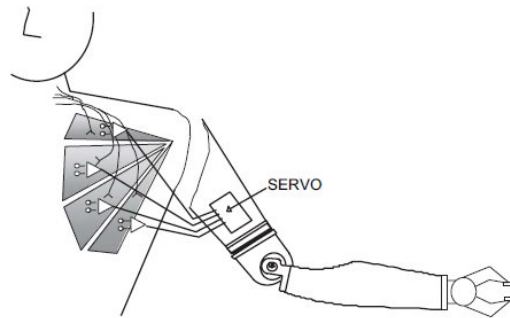
# 上肢假肢控制——靶向肌肉神经再支配 (TMR)

## Upper-limb Prosthetic Control

### Targeted Muscle Reinnervation

- 目标：仍用EMG作为输入，但通过"重新接线"运动神经到完好肌肉，实现更自然的假肢控制

Goal is to create more natural control of prosthetic limb while still using EMG as input by "re-wiring" motor neurons to intact muscle



1. Denervate an "expendable" area of muscle, such as in the chest.
2. Attach motor neurons from the amputated arm to different areas of the denervated chest muscles.
3. Use surface electrodes over chest muscles to provide many channels of EMG control input to prosthetic.

**If reinnervation is successful, thinking about using the arm would result in contraction of the reinnervated muscles.**

• 步骤：

1. 去神经化一块"可有可无"的肌肉区域，如胸部肌肉
2. 把截肢手臂的运动神经接到这些去神经化的胸部肌肉上
3. 在胸部肌肉上贴表面电极，提供多路EMG控制信号

• 关键效果：如果再支配成功，患者"想到"用手臂，就会引起这些再支配肌肉的收缩

TMR是Kuiken医生发明的经典手术——截肢后，原来控制手指的神经没有地方去了，就把它们接到胸大肌上。患者想"张开手"，胸肌就会动，贴在胸肌上的电极就能采到这个信号。相当于"借"胸肌当信号放大器。

# 外周神经控制 (Peripheral Nerve Control)

## Upper-limb Prosthetic Control

### Peripheral Nerve Control

Use of peripheral nerves as control input rather than EMG

Like TMR, an attempt to make limb control more intuitive than “coded” EMG

Sensory feedback can be sent through peripheral nerves, giving the user more control over joint position and grip force

Pulse train of varying current amplitude sent to electrodes in sensory nerves to re-create signals conveying touch and proprioception

Motor neuron firing rate decoded as force or torque command

- 用外周神经而非EMG作为控制输入
- 和TMR类似，试图让肢体控制比“编码式”EMG更直觉
- 感觉反馈可以通过外周神经传回，让用户获得关节位置和握力的感知
- 向感觉神经发送不同电流幅度的脉冲串，重建触觉和本体感觉信号
- 运动神经元放电率被解码为力或力矩指令



#### Achievements:

- Sensory information has potential to put human back in the loop during grasping
- Interpreting firing rate as torque or force command rather than trying to classify as a set motion

#### Shortcomings:

- Procedure involves implant to interface nerves to controller
- Pilot study was for single-DOF scenarios (grip force and elbow angle)

#### 成就:

- 感觉信息有潜力在抓握过程中把人带回控制回路
- 把放电率解读为力矩或力指令，而非分类为某个固定动作

#### 不足:

- 需要手术植入神经-控制器接口
- 试点研究仅针对单自由度场景（握力和肘关节角度）

Cyberhand是欧洲的标志性项目，目标是真正"神经接入"的假手——不仅能读大脑/神经信号控制假手，还能把假手上的触觉传感器信号传回感觉神经，形成闭环。

# Upper-limb Prosthetic Control

## 控制论假肢

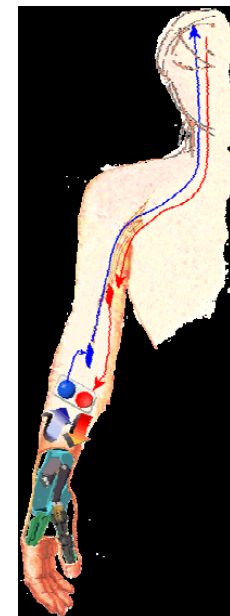
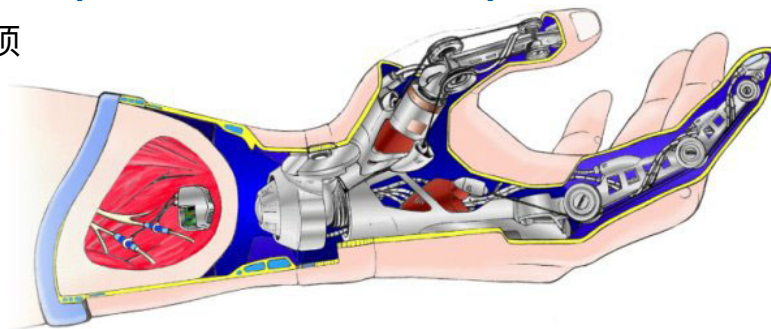
Cybernetic prosthesis

(from the EU-FET "CYBERHAND" Project):

a cybernetic prosthesis controlled by the brain

来自欧盟FET "CYBERHAND"项目

目：由大脑控制的控制论假肢



Neurobotics /prostheses : interfaces between an assistive device and the human nervous system such that the user's brain functions become part of the system control loop

R&D issues

- biocompatible implantable micro-sensors and electrodes
- Neurophysiology: which neurons to interface?
- Data processing (volitional command and artificial sensory feedback)...

- 神经机器人学/假肢：辅助装置与人神经系统之间的接口，使用户的大脑功能成为系统控制回路的一部分
- 研发问题：
  - 生物相容性可植入微传感器和电极
  - 神经生理学：该与哪些神经元接口？
  - 数据处理（随意运动指令和人工感觉反馈） .....

# CyberHand架构

## Upper-limb Prosthetic Control

- CyberHand Architecture

3. 再生型电极（传出神经）：  
读运动指令

4. 再生型电极（传入神经）：  
传感觉反馈

8. 解码患者  
意图，人工  
手的嵌入式  
闭环控制

### Regeneration-type electrodes:

- 3. Regeneration-type electrode (efferent nerve)
- 4. Regeneration-type electrode (afferent nerve)

### 8. Decoding patient's intentions and Embedded closed-loop control of the artificial hand

1. 生物机电手

### 1. Biomechatronic Hand

Stump

### 5. Implanted neural interface:

- ◆ ENG efferent signals recording (patient's intention detection)
- ◆ Afferent nerves stimulation (to provide sensory feedback to the patient)

5. 植入式神经接口：

- ENG传出信号记录（检测患者意图）
- 传入神经刺激（向患者提供感觉反馈）

6. 接收器

6. Receiver

7. Transmitter

7. 发射器

### 2. Embedded Biomimetic sensors:

- within the structure
- within the glove

2. 嵌入式仿生传感器：  
在结构内  
在手套内

这是完整的双向闭环——假手传感器感知到的力，通过植入电极直接刺激患者的感觉神经，患者“感觉到”假手碰到了东西。



# Cybernetic prosthesis

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Scuola Superiore  
Sant'Anna

di Studi Universitari e di Perfezionamento

**ARTS** Lab

Advanced Robotics Technology and Systems

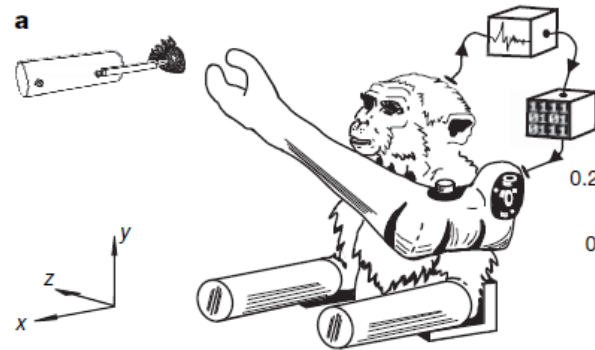
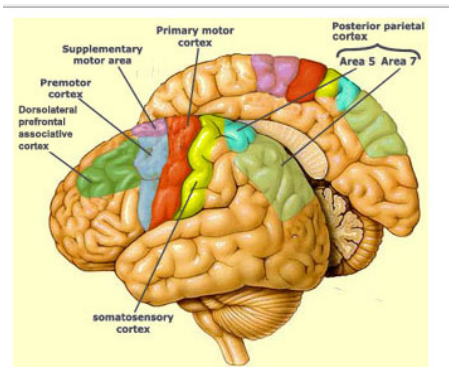


# 上肢假肢控制——皮层控制

## Upper-limb Prosthetic Control

### Cortical Control

- Upper limb motion representation in the brain:
  - Certain region of primary motor cortex associated with arm movement
  - Speed and direction of desired motion can be inferred from neural activity
  - Population Vector Hypothesis (Georgopoulos 1988)
  - Each neuron has a preferred firing direction; activity of that neuron indicates motion in preferred direction.
  - Represent velocity of hand as vector sum of preferred firing directions weighted by firing rate



这是脑机接口（BCI）的理论基础——不是单个神经元对应一个动作，而是一群神经元的活动方向加权求和，就能算出“手想往哪走、走多快”。

- 大脑中上肢运动的表征：
  - 初级运动皮层的特定区域与手臂运动相关
  - 期望运动的速度和方向可以从神经活动中推断
  - 群体向量假说（Population Vector Hypothesis, Georgopoulos 1988）：
    - 每个神经元有一个“偏好放电方向”；该神经元的活动表示沿偏好方向的运动
    - 手的速度表示为所有神经元偏好方向按放电率加权的向量和



# 假肢装置的先进技术

## Advanced Technologies for Prosthetic Devices

### 神经工程 Neuroengineering

Neuroengineering is defined as the interdisciplinary field of engineering and computational approaches to problems in basic and clinical neurosciences.

Neuroengineering encompasses the fields of

Engineering

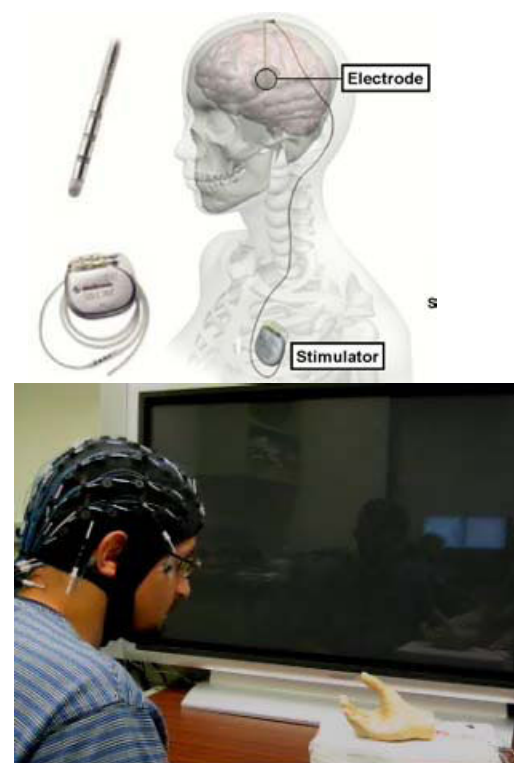
Mathematics

Computer science

Molecular and cellular systems

Neurosciences

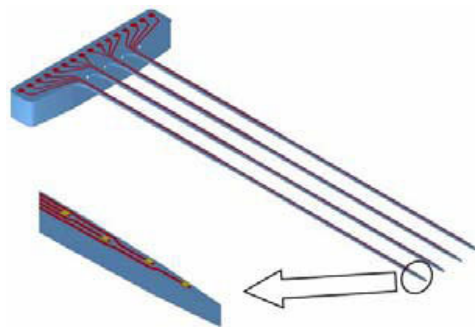
- 神经工程：用工程和计算方法解决基础与临床神经科学问题的交叉学科
- 涵盖领域：工程、数学、计算机科学、分子与细胞系统、神经科学



# 神经假体的步骤：植入式微电极阵列

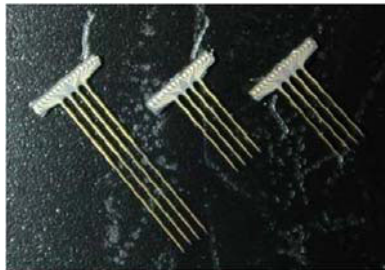
## Advanced Technologies for Prosthetic Devices

### Steps to Neural Prosthesis: Implanted Microelectrode Arrays



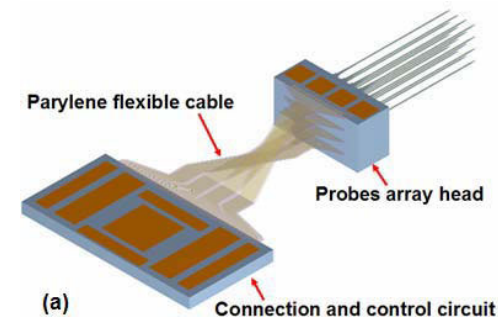
- 每个探针杆上有4个触点

Each shank on each probe has four contacts.



Under Development at U of Michigan, U of Utah and Cal Tech

- 密歇根大学、犹他大学、加州理工在开发



- 通过Parylene柔性电缆连接到控制电路

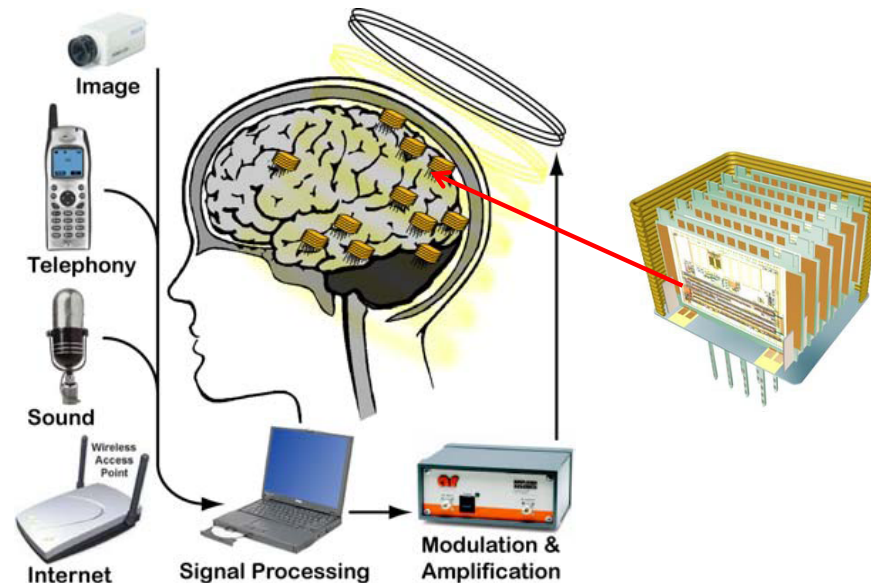
Multiple probes form a 3-D geometry

- 多个探针形成3D结构

# Advanced Technologies for Prosthetic Devices

## Steps to Neural Prosthesis: A Distributed Wireless 3-D Implants

神经假体的步骤：分布式  
无线3D植入



- 概念：多个小型植入器分散在大脑各处，无线供电和通信
- 应用：图像、电话、声音、互联网——脑机接口同时恢复多种感官/沟通能力
- 流程：植入器采集信号 → 调制放大 → 信号处理 → 输出到外部设备

# Neuralink



- [https://www.youtube.com/watch?v=Gv\\_XB6Hf6gM](https://www.youtube.com/watch?v=Gv_XB6Hf6gM)

图片：Neuralink在患者头骨上植入的接口座（Wired杂志报道）

# 假肢装置的先进技术（现状 vs 未来）

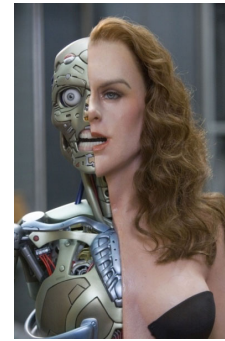
## Advanced Technologies for Prosthetic Devices

Current



- Current（当前）：
  - 身上挂着粗大的外接机械臂管线
  - 患者戴着外骨骼助行器在医院训练

Future



- Future（未来）：
  - 电影里的脑控全息操作台
  - 终结者式的完全拟人化机器人

# 辅助与康复机器人技术的未来

## Future of Assistive and Rehabilitation Robotics Technologies



### Disability is still prevalent, particularly with ageing population

Engineering disciplines permeate many aspects of rehabilitation:

**ECE** (Signal processing, control/information theory, neural networks, micro-fabrication, fuzzy logic, virtual reality, image processing...), **BME** (Biomechanical testing, gait/motion analysis, prosthetic fitting, seat cushion and back support design, and design of artificial joints), **Material E** (Newer biocompatible materials), **Chemical E** (Implantable sensors), **Industrial E** (rehabilitative ergonomics)

**RE challenge is to find advances in any field, engineering or otherwise, that aid disabled clients to function independently with skills to integrate into society**

- 残疾仍然普遍存在，尤其是在老龄化人口中
- 工程学科渗透康复的方方面面：
  - ECE（电子计算机工程）：信号处理、控制/信息论、神经网络、微制造、模糊逻辑、虚拟现实、图像处理
  - BME（生物医学工程）：生物力学测试、步态/运动分析、假肢装配、坐垫/靠背设计、人工关节设计
  - 材料工程：新型生物相容材料 ◦ 化学工程：可植入传感器 ◦ 工业工程：康复人因工程
- 康复工程（RE）的挑战：在任何领域（工程或其他）找到进展，帮助残障客户独立生活、掌握融入社会的技能



## Course Focused Topics (Part 2 by Prof. Yu)

- Stroke and Stroke Rehabilitation
- Robotics for upper limb rehabilitation
- Robotics for upper lower rehabilitation

标题：课程重点主题（第二部分）

1. 中风与中风康复
2. 上肢康复机器人
3. 下肢康复机器人



# Summary

- Driving forces of Robotics in Healthcare
- Overview of various rehab and assistive robotics
- Outline of course focused topics (Part 2)

标题：总结

- 医疗机器人的驱动力
- 各类康复与辅助机器人概述
- 第二部分课程重点主题大纲

